Band-1 Hadamard order 14356 – Classical Paley-II DIRECT (q2=7177); IN SPK via Paley-II; Gap 2052 Still OPEN

H_14356 Paley-II PASS; Paley-I FAIL/SKIP (q=14355=3^2*5*11*29 *NOT prime); Kronecker FAIL/SKIP (4*3589; 3589=37*97 odd NOT Hadamard-admissible); Miyamoto digression skipped; gap 2052 OPEN; conjecture OPEN; GATE-04 LOCKED

C.M. Baird; ZoraASI mission-control Band-1 dash

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Document class. Scientific computational / construction certificate (Band-1).
Program. Open-mathematics collapse list * Human+AI attack order * ZoraASI mission-control.
Parents. `2026-10-09_Band1_Hadamard_14348_PaleyI` (chain scaffolding); `2026-09-21_Band1_Hadamard_9124_P` (Paley-II construction template); `2026-10-09_Band1_Hadamard_14328_PaleyI`; `2026-10-09_Band1_Hadamard_143` `2026-10-09_Band1_Hadamard_14304_PaleyI`; ....
Lab. `/workspace/experiments/band1/hadamard_14356_paleyII/`.
Cutoff. 2026-10-09T17:22:13Z (~1:22PM ET).

Honesty stamp (non-negotiable).
This memo claims **finite construction of order 14356 only (verified max|HH^T-nI|=0), via classical Paley-II q2=7177 (prime ===1 mod 4), order 2(q2+1). 14356 IN SPK via Paley-II. 14356 is a long-known Hadamard order; this is a finite on-box verification, NOT a new-order claim.** Honest skip: Miyamoto digression this wave (prefer classical SPK). Classical SPK for 14356: Paley-II **PASS** ($q_2 = 7177$ prime $\equiv 1 \pmod 4$, $n = 2(q_2 + 1)$); Paley-I **FAIL/SKIP** ($q = 14355 = 3^2 \cdot 5 \cdot 11 \cdot 29$ NOT prime); Kronecker **FAIL/SKIP** ($14356 = 4 \cdot 3589$; $3589 = 37 \cdot 97$ odd, NOT Hadamard-admissible; do NOT invent H_3589). **14356** folds into SPK **only** via classical Paley-II (this run). It **does not** claim the full Hadamard conjecture. GATE-04 collection remains **LOCKED** (no invented J). No finished-ToE claim. No RH / P≠NP / YM / Frankl / NS-solved / AC-solved claims. Gap **2052** remains **OPEN**. `clay_closed = false`. Prior SPK **14348+14328+14324+14304+14252+14244+14208+14160+14144+14108+14088+14084+14072+140** retained; Miyamoto-only **8948/8644/8516/8452/7988/...** excluded.

Abstract

Finite construction of order **14356** (first Paley-II DIRECT wave since 9124).

- H_{14356} = Paley-II ($q_2 = 7177$) – classical SPK
- Paley-I for 14356: **FAIL/SKIP** ($q = 14355 = 3^2 \cdot 5 \cdot 11 \cdot 29$ NOT prime)
- Kronecker for 14356: **FAIL/SKIP** ($4 \cdot 3589$; $3589 = 37 \cdot 97$ odd, NOT Hadamard-admissible)
- Skip: Miyamoto digression this wave

SPK: IN SPK via Paley-II.
Gap 2052: OPEN. **Conjecture:** OPEN. **clay_closed:** false. **GATE-04:** LOCKED.

`python3 verify_hadamard_14356.py`

Results. overall_pass=True; units 719/719; row blocks 113/113 (block size 128); wall 2664.6209s; constructed max|HHˆT-nI|=0.

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Arithmetic

- Target $n = 14356 = 2(q_2 + 1)$ with $q_2 = 7177$ prime $\equiv 1 \pmod 4$ — Paley-II eligible (claim path).
- Paley-I: $q = n - 1 = 14355 = 3^2 \cdot 5 \cdot 11 \cdot 29$ NOT prime — FAIL/SKIP.
- Kronecker: $14356 = 4 \cdot 3589$; $3589 = 37 \cdot 97$ odd ($\equiv 1 \pmod 4$) NOT Hadamard-admissible — FAIL/SKIP; no other admissible split.
- Shape: $14356 = 2^2 \cdot 37 \cdot 97 \equiv 4 \pmod{16}$.
- Miyamoto digression skipped this wave (prefer classical SPK).

Construction (Paley type II)

Let χ be the quadratic character of $\mathbb{F}_{7177}$ and $Q_{ij} = \chi(j - i)$ (symmetric, since $q_2 \equiv 1 \pmod 4$). The conference matrix $C = \begin{pmatrix} 0 & \mathbf{1}^T \\ \mathbf{1} & Q \end{pmatrix}$ of order $q_2 + 1 = 7178$ gives

$$H_{14356} = \begin{pmatrix} C + I & C - I \\ C - I & -(C + I) \end{pmatrix}.$$

Built directly in int8 (memory-lean; unit-tested equal to the int64 reference builder on $q \in \{5, 13, 17, 29, 37, 41, 53, 61\}$), then verified block-wise: every row block of $HH^T - nI$ checked exactly in int32.

Constructions (verified)

Object	Method	max|HHˆT-nI|	Wall (s)
H_14356 Paley-II	Paley-II q2=7177	0	2645.3581
H_14356 Paley-I	FAIL/SKIP (q=14355=3ˆ2*5*11*29 NOT prime)	–	–
H_14356 Kronecker	FAIL/SKIP (4*3589; 3589=37*97 odd NOT Hadamard-admissible)	–	–

SPK closure note

Quantity	Value
14356 in SPK	**True** (Paley-II q2=7177)
Prior 14348	IN SPK via Paley-I (~12:01PM ET 2026-10-09)
Gap 2052	**OPEN**
clay_closed	false
GATE-04	LOCKED

Honesty

- clay_closed = false
- GATE-04 LOCKED (no invented J)
- No finished-ToE claim
- Hadamard conjecture OPEN
- Gap 2052 OPEN (method triage / non-finding only)
- No RH / P!=NP / YM / Frankl / NS / AC solved claims
- 14356 is a long-known order (Paley 1933 family); finite on-box verification only
- SPK membership only via classical Paley-I/II/Kronecker (not Miyamoto alone)

Next optional

NEXT direct route after 14356: q=14387->n=14388 Paley-I DIRECT (q=14387 prime ===3 mod 4; only other prime in 14356..14386 is 14369 ===1 mod 4 -> skipped; Paley-II q2=7193 prime ===1 mod 4 -> ALSO a direct route (deferred note-only; Paley-I is the claim path, as 14244); Kronecker 14388=4*3597; 3597=3*11*109 odd (===1 mod 4) -> NOT Hadamard-admissible -> FAIL/SKIP (do NOT invent H_3597); 14388=2ˆ2*3*11*109 ===4 mod 16) | no direct route in 14360..14384

(sympy scan) and no Kronecker-only route there (14368=4*3592, 14384=4*3596 admissible but H_m has no direct classical Paley route: 3591=3^3*719, 1795=5*359; 3595=5*719, 1797=3*599) | skipped Paley-II DIRECT orders below 14356 still unbuilt (each Paley-I FAIL, Kronecker odd-m FAIL/SKIP; template now hadamard_14356_paleyII): 13924 (q2=6961) / 13956 (q2=6977) / 13996 (q2=6997) / 14004 (q2=7001) / 14028 (q2=7013) / 14116 (q2=7057) / 14140 (q2=7069) / 14220 (q2=7109) / 14260 (q2=7129) (+13900 q2=6949 historically not listed)